Private Access (ZPA)

Understanding the Log Stream Content Format

The following information includes specifications and guidelines about the log field format that is used by these log types:

- Audit Logs
- App Connector Metrics
- App Connector Status
- Browser Access
- Microsegmentation Flow
- Private Cloud Controller Metrics
- Private Cloud Controller Status
- Private Service Edge Metrics
- Private Service Edge Status
- User Activity
- User Status
- Web Inspection

While configuring your log receiver, you can edit the default Log Stream Content to include customized log fields. For example, when configuring your log receiver using the App Connector Status log type, the `ConnectionLogType` field can be added as a custom log field to distinguish between AppProtection and event logs. The expected values for this field are `event_log` and `inspection_log`. The supported log field format specification must be included (i.e., %[OPT]s, %[OPT]j, %[OPT]J, %[OPT]d, %[OPT]x, %[OPT]f, %[OPT]o).

Log Field Format Specifications

The field format string contains arbitrary text with the following escape formats, where [OPT] is an optional field that determines the width/precision of the style, implemented by printf. You can also specify the padding if you want constant width strings. For example:

- `%d{LogTimestamp:hh}:%d{LogTimestamp:mm}:%d{LogTimestamp:ss}` would appear as `21:5:0`
- `%02d{LogTimestamp:hh}:%02d{LogTimestamp:mm}:%02d{LogTimestamp:ss}` would appear as `21:05:00`

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The <field_name> in the following table is substituted by the corresponding value when the log receiver sends logs to the SIEM.

Field Format Specification	Description
%[OPT]s{<field_name>}	Displays the field as a string. For example, `User is %s{user_id}, and time is %s{startrx_us:iso8601}` would appear as `User is john.doe, and time is 2017-09-05T09:44:06.363Z`.
%[OPT]s{LogTimestamp:<field_name>}	Displays the LogTimestamp field as a string. You can use the following special characters for timestamps (in epoch): • us = microseconds (0 to 999999) • ms = milliseconds (0 to 999) • ss = seconds (0 to 59) • mm = minutes (0 to 59) • hh = hours (0 to 23) • dd = day of month (1 to 31) • mth = month of year (1 to 12) • mon = name of month (3 characters) • yyy = year (3 digits) • day = day of week (3 characters) • time = example: Thu Sep 6 22:55:48 2012 • iso8601 = example: 2017-09-05T09:44:06.363Z • epoch = epoch in seconds • epoch_ms = epoch in milliseconds • epoch_us = epoch in microseconds For example, `%s{LogTimestamp:time}` would appear as `Mon Sep 10 10:50:46 2012`
%[OPT]j{<field_name>}	Displays the field as a JSON string with quotation marks

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Field Format Specification	Description
%[OPT]J{<field_name>}	Displays the field as a JSON string without quotation marks
%[OPT]d{<field_name>}	Displays the field in decimal integer format
%[OPT]x{<field_name>}	Displays the field as a number in hexadecimal (base-16) format
%[OPT]f{<field_name>}	Displays the field as a float
%[OPT]o{<field_name>}	Displays the field in a 256-bit (64 character) one-way hashed (obfuscated) form. If you want a smaller obfuscation string, you can use precision limiting.

Log Field Format Guidelines

The applicability of the following log field format guidelines and recommendations might vary based on the capabilities of your SIEM:

- Fields must be in the order you want them to appear in the output. If your SIEM can support it, consider using name-value pairs. Some SIEMs, such as Splunk, can automatically parse name-value formats and auto-detect field names regardless of the order. To learn more about third-party SIEM integrations, see Private Access and Splunk Deployment Guide and the Zscaler and Splunk Deployment Guide.
- Field names are case sensitive.
- The following special characters can be used generate control outputs:
  - \t = tab
  - \n = newline
  - \r = carriage return
- The default Log Stream Content automatically adds a newline (\n) character after the last field. If you have modified the content, you must append at least one newline character after the last field.
- Optionally, you can include static text strings in the output. For example, `User is %s{user_id}, and time is %s{LogTimestamp:time} GMT` would appear as `User is john.doe, and time is Mon Sep 10 10:50:46 2012 GMT`

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